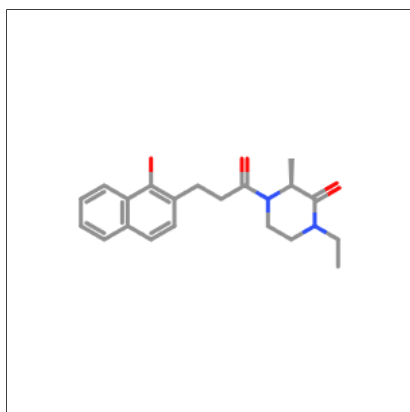
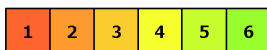


## Oral toxicity prediction results for input compound



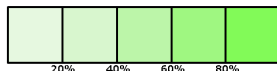
Predicted LD50: 1500mg/kg

Predicted Toxicity Class: 4



Average similarity: 71.02%

Prediction accuracy: 69.26%



|                                           |                                   |
|-------------------------------------------|-----------------------------------|
| Name                                      | CCN1CCN(C(=O)CCc2cc[C@@H](C)C1=O) |
| Molweight                                 | 340.42                            |
| Number of hydrogen bond acceptors         | 5                                 |
| Number of hydrogen bond donors            | 1                                 |
| Number of atoms                           | 25                                |
| Number of bonds                           | 27                                |
| Number of rotatable bonds                 | 5                                 |
| Molecular refractivity                    | 106.11                            |
| Topological Polar Surface Area            | 60.85                             |
| octanol/water partition coefficient(logP) | 2.43                              |

## Toxicity Model Report

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| Classification                             | Target                                                                                | Shorthand     | Prediction | Probability |
|--------------------------------------------|---------------------------------------------------------------------------------------|---------------|------------|-------------|
| Organ toxicity                             | Hepatotoxicity                                                                        | dili          | Inactive   | 0.87        |
| Organ toxicity                             | Neurotoxicity                                                                         | neuro         | Active     | 0.63        |
| Organ toxicity                             | Nephrotoxicity                                                                        | nephro        | Inactive   | 0.62        |
| Organ toxicity                             | Respiratory toxicity                                                                  | respi         | Active     | 0.84        |
| Organ toxicity                             | Cardiotoxicity                                                                        | cardio        | Inactive   | 0.75        |
| Toxicity end points                        | Carcinogenicity                                                                       | carcino       | Inactive   | 0.68        |
| Toxicity end points                        | Immunotoxicity                                                                        | immuno        | Active     | 0.72        |
| Toxicity end points                        | Mutagenicity                                                                          | mutagen       | Inactive   | 0.67        |
| Toxicity end points                        | Cytotoxicity                                                                          | cyto          | Inactive   | 0.66        |
| Toxicity end points                        | BBB-barrier                                                                           | bbb           | Active     | 0.58        |
| Toxicity end points                        | Ecotoxicity                                                                           | eco           | Inactive   | 0.59        |
| Toxicity end points                        | Clinical toxicity                                                                     | clinical      | Active     | 0.72        |
| Toxicity end points                        | Nutritional toxicity                                                                  | nutri         | Inactive   | 0.62        |
| Tox21-Nuclear receptor signalling pathways | Aryl hydrocarbon Receptor (AhR)                                                       | nr_ahr        | Inactive   | 0.75        |
| Tox21-Nuclear receptor signalling pathways | Androgen Receptor (AR)                                                                | nr_ar         | Inactive   | 0.97        |
| Tox21-Nuclear receptor signalling pathways | Androgen Receptor Ligand Binding Domain (AR-LBD)                                      | nr_ar_lbd     | Inactive   | 0.98        |
| Tox21-Nuclear receptor signalling pathways | Aromatase                                                                             | nr_aromatase  | Inactive   | 0.91        |
| Tox21-Nuclear receptor signalling pathways | Estrogen Receptor Alpha (ER)                                                          | nr_er         | Inactive   | 0.89        |
| Tox21-Nuclear receptor signalling pathways | Estrogen Receptor Ligand Binding Domain (ER-LBD)                                      | nr_er_lbd     | Inactive   | 0.98        |
| Tox21-Nuclear receptor signalling pathways | Peroxisome Proliferator Activated Receptor Gamma (PPAR-Gamma)                         | nr_ppar_gamma | Inactive   | 0.98        |
| Tox21-Stress response pathways             | Nuclear factor (erythroid-derived 2)-like 2/antioxidant responsive element (nrf2/ARE) | sr_are        | Inactive   | 0.96        |
| Tox21-Stress response pathways             | Heat shock factor response element (HSE)                                              | sr_hse        | Inactive   | 0.96        |
| Tox21-Stress response pathways             | Mitochondrial Membrane Potential (MMP)                                                | sr_mmp        | Inactive   | 0.87        |
| Tox21-Stress response pathways             | Phosphoprotein (Tumor Suppressor) p53                                                 | sr_p53        | Inactive   | 0.88        |
| Tox21-Stress response pathways             | ATPase family AAA domain-containing protein 5 (ATAD5)                                 | sr_atad5      | Inactive   | 0.98        |
| Molecular Initiating Events                | Thyroid hormone receptor alpha (THRα)                                                 | mie_thr_alpha | Inactive   | 0.87        |
| Molecular Initiating Events                | Thyroid hormone receptor beta (THRβ)                                                  | mie_thr_beta  | Inactive   | 0.54        |
| Molecular Initiating Events                | Transthyretin (TTR)                                                                   | mie_ttr       | Inactive   | 0.96        |
| Molecular Initiating Events                | Ryanodine receptor (RYR)                                                              | mie_ryr       | Inactive   | 0.78        |

| Classification              | Target                                                                      | Shorthand  | Prediction | Probability |
|-----------------------------|-----------------------------------------------------------------------------|------------|------------|-------------|
| Molecular Initiating Events | <u>GABA receptor (GABAR)</u>                                                | mie_gabar  | Inactive   | 0.54        |
| Molecular Initiating Events | <u>Glutamate N-methyl-D-aspartate receptor (NMDAR)</u>                      | mie_nmdar  | Inactive   | 0.88        |
| Molecular Initiating Events | <u>alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate receptor (AMPA)</u> | mie_ampar  | Inactive   | 1.0         |
| Molecular Initiating Events | <u>Kainate receptor (KAR)</u>                                               | mie_kar    | Inactive   | 1.0         |
| Molecular Initiating Events | <u>Achetylcholinesterase (AChE)</u>                                         | mie_ache   | Active     | 0.50        |
| Molecular Initiating Events | <u>Constitutive androstane receptor (CAR)</u>                               | mie_car    | Inactive   | 0.99        |
| Molecular Initiating Events | <u>Pregnane X receptor (PXR)</u>                                            | mie_pxr    | Inactive   | 0.81        |
| Molecular Initiating Events | <u>NADH-quinone oxidoreductase (NADHox)</u>                                 | mie_nadhox | Inactive   | 0.94        |
| Molecular Initiating Events | <u>Voltage gated sodium channel (VGSC)</u>                                  | mie_vgsc   | Inactive   | 0.80        |
| Molecular Initiating Events | <u>Na<sup>+</sup>/I<sup>-</sup> symporter (NIS)</u>                         | mie_nis    | Inactive   | 0.97        |
| Metabolism                  | <u>Cytochrome CYP1A2</u>                                                    | CYP1A2     | Inactive   | 0.91        |
| Metabolism                  | <u>Cytochrome CYP2C19</u>                                                   | CYP2C19    | Inactive   | 0.80        |
| Metabolism                  | <u>Cytochrome CYP2C9</u>                                                    | CYP2C9     | Inactive   | 0.58        |
| Metabolism                  | <u>Cytochrome CYP2D6</u>                                                    | CYP2D6     | Inactive   | 0.57        |
| Metabolism                  | <u>Cytochrome CYP3A4</u>                                                    | CYP3A4     | Inactive   | 0.70        |
| Metabolism                  | <u>Cytochrome CYP2E1</u>                                                    | CYP2E1     | Inactive   | 1.0         |

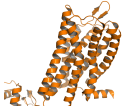
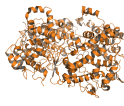
### Toxicity targets

Possible binding to toxicity targets is shown below. For more information on the targets, please click on the individual abbreviations.



| AA2AR | ADRB2 | ANDR | AOFA | CRFR1 | DRD3 | ESR1 | ESR2 | GCR | HRH1 | NR1I2 | OPRK | OPRM | PDE4D | PGH1 | PRGR |
|-------|-------|------|------|-------|------|------|------|-----|------|-------|------|------|-------|------|------|
|       |       |      |      |       |      |      |      |     |      |       |      |      |       |      |      |

Details about possible toxicity targets:

|                                                                                     | Toxicity Target              | Avg Pharmacophore Fit | Avg Similarity Known Ligands |
|-------------------------------------------------------------------------------------|------------------------------|-----------------------|------------------------------|
|  | Amine Oxidase A              | 61.04%                | 0%                           |
|  | Opioid Receptor Mu           | 0%                    | 74.08%                       |
|  | Prostaglandin G/H Synthase 1 | 37.74%                | 0%                           |